



BantayLaot: Fish Catch Documentation and IUU Fishing Monitoring for Municipal Fisheries in the Philippines Using Mobile and Web Technologies

Jose Benjamin Gallero and Dr. Val Randolph M. Madrid



- Rich maritime heritage of the Philippines
- **Challenges:** IUU fishing threatens sustainability
- Economic and social impact on coastal communities



Sector	Illegal	Unregistered	Unreported
Municipal	257,000-402,000 MT/year P24.1-37.8 billion	80,000-125,000 unregistered municipal fishing vessels	309,000-485,000 MT/year P29.0-45.6 billion
Commercial	259,000-364,000 MT/year P17.7-17.8 billion	1,600-2,700 unregistered or incorrectly registered commercial fishing vessels	274,000-422,000 MT/year P18.7-28.7 billion
Total	516,000-766,000 MT/year P41.8-62.6 billion	81,600-127,700 unregistered fishing vessels	583,000-907,000 MT/year P47.7-74.3 billion

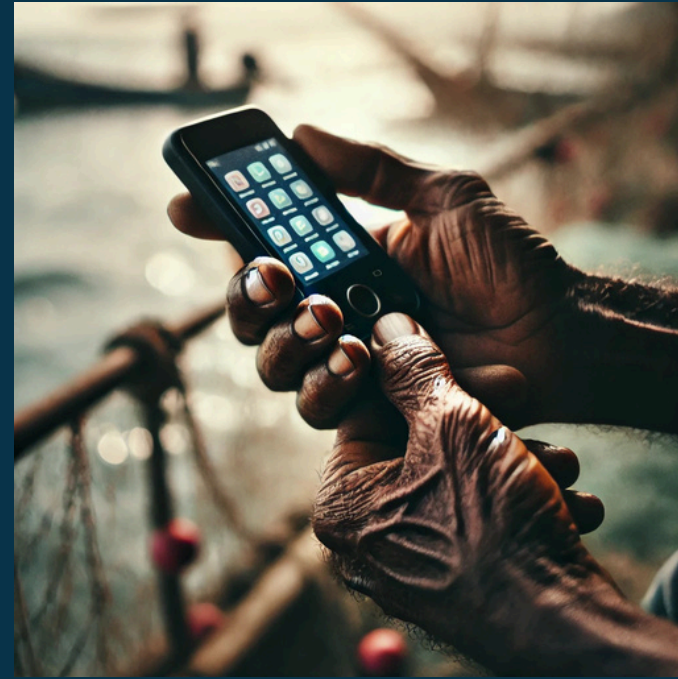
Source: Quantifying the Prevalence and Impact of Illegal, Unreported, and Unregulated Fishing in the Philippines, Survey and Workshop Report, 31 January 2021

- ₱62 billion annual losses
- 30% unregistered
- 422,000 metric tons
- 30.6% Poverty Rates





**Vessel Tracking
Feature**



**Fish Catch
Documentation**



**IUU Reporting
Feature**

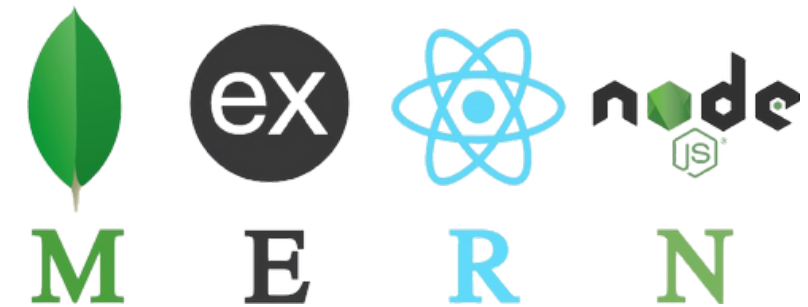


**Data
Visualisation**

- Transparency and accountability
- Sustainable fishing practices
- Data-driven decision-making
- Benefits: fisherfolk, government, researchers



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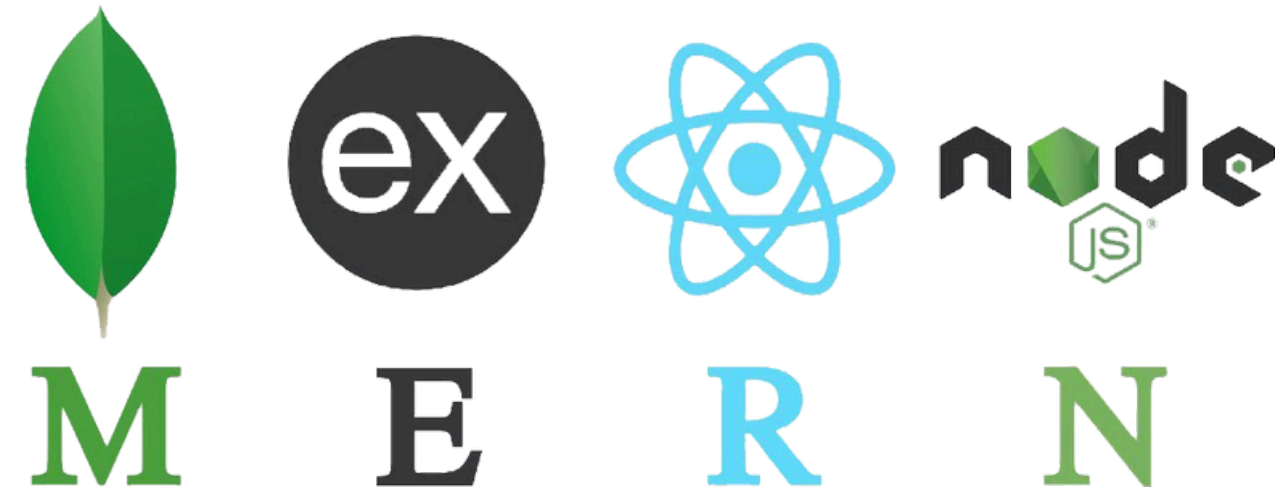
- **Technology:** Kotlin (Mobile), MERN Stack (Web)
- **Development Tools:** Android Studio, Visual Studio Code
- **System Components:** Mobile App & Web Platform

- Kotlin
- Jetpack
- Compose Room
- **Features:**
 - GPS Tracking
 - Offline Mode
 - Violation Reporting



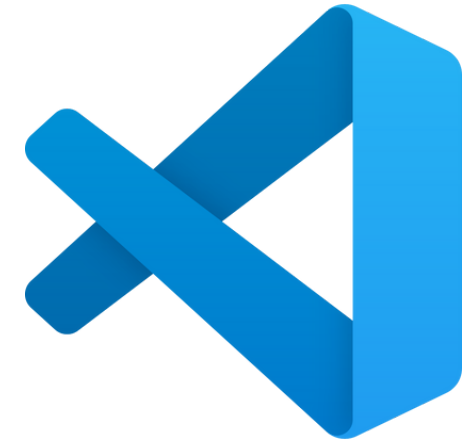
- **Stack:**

- MongoDB
- Express.js
- React
- Node.js



- **Features:**

- Geospatial Visualization
- CPUE Analysis
- Data Filtering

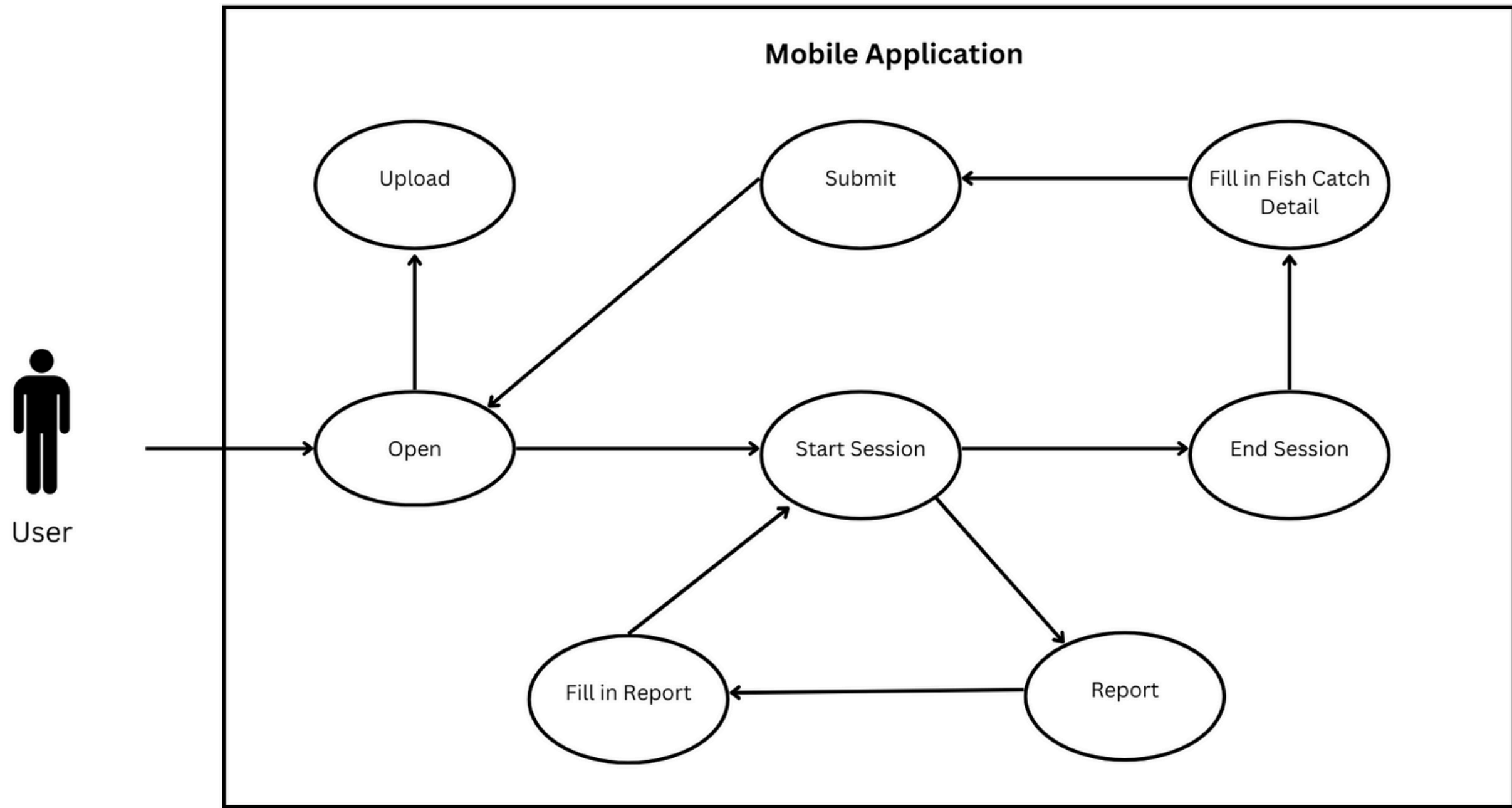


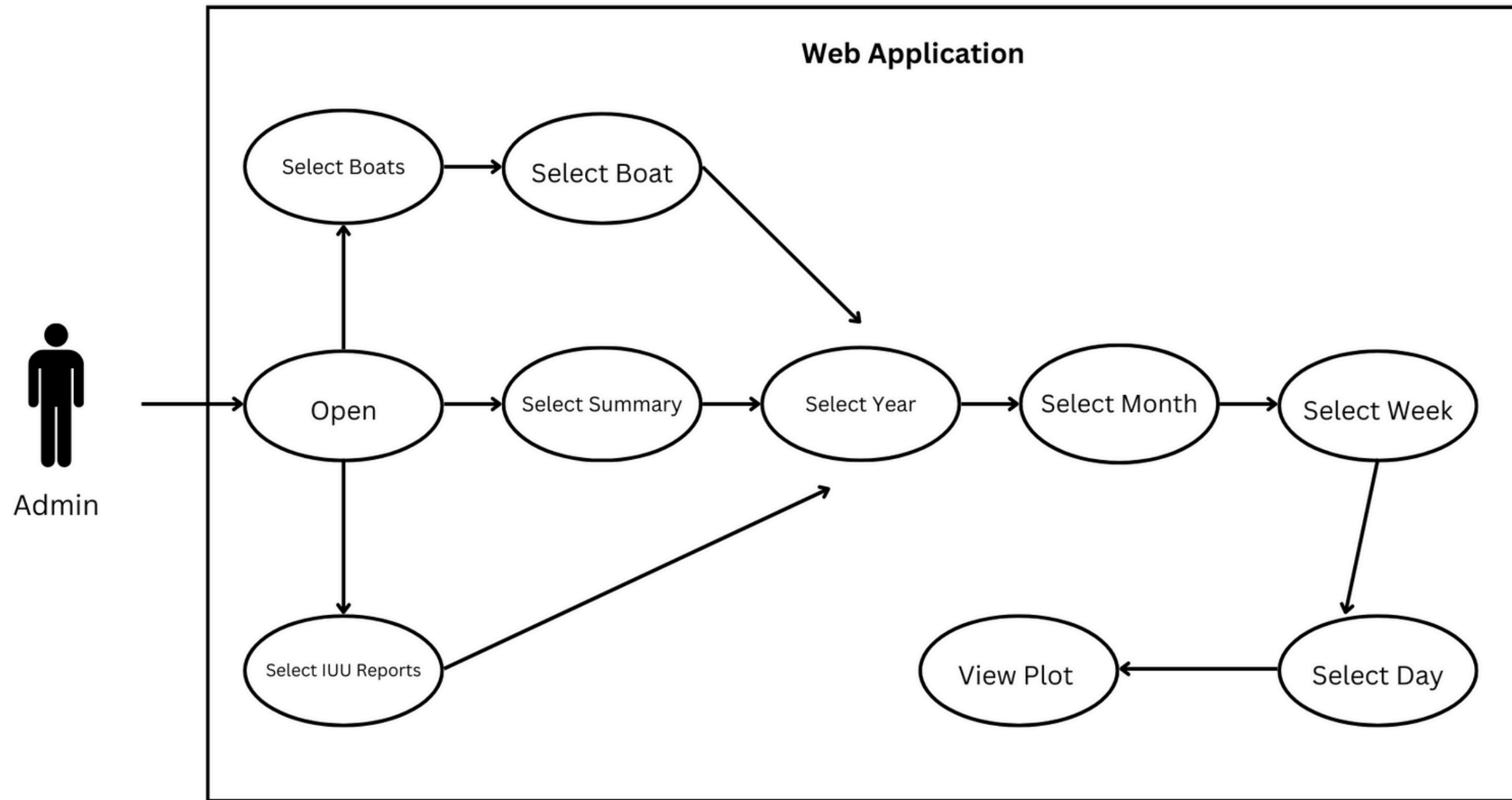
- **Session Data:**

- Fishing Gear: Types used during sessions
 - Fish Catch Information: Species and catch weights for each gear
- Location Tracking: GPS coordinates
- Session Details: Start and end timestamps
- IUU Reporting Data

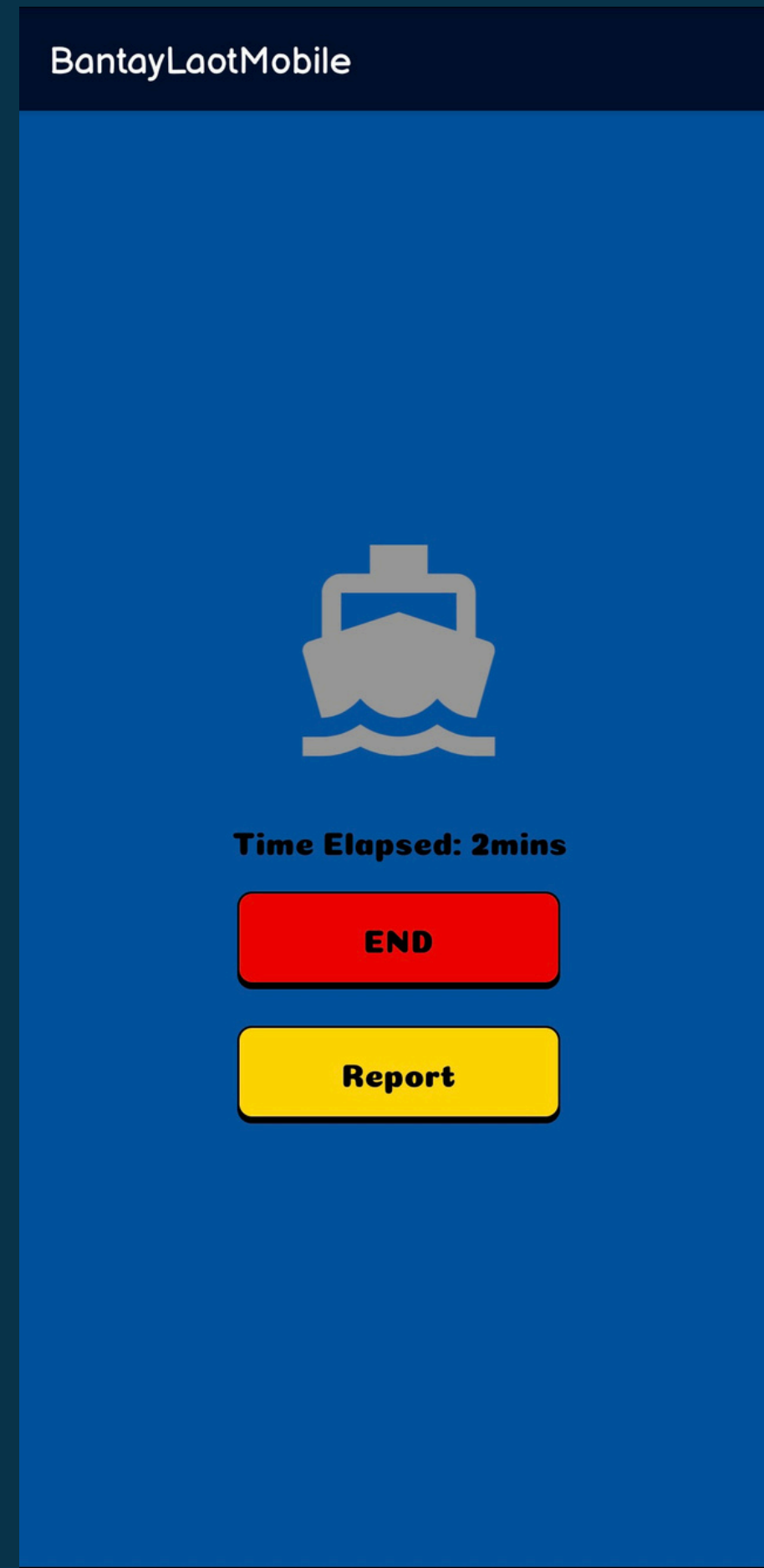
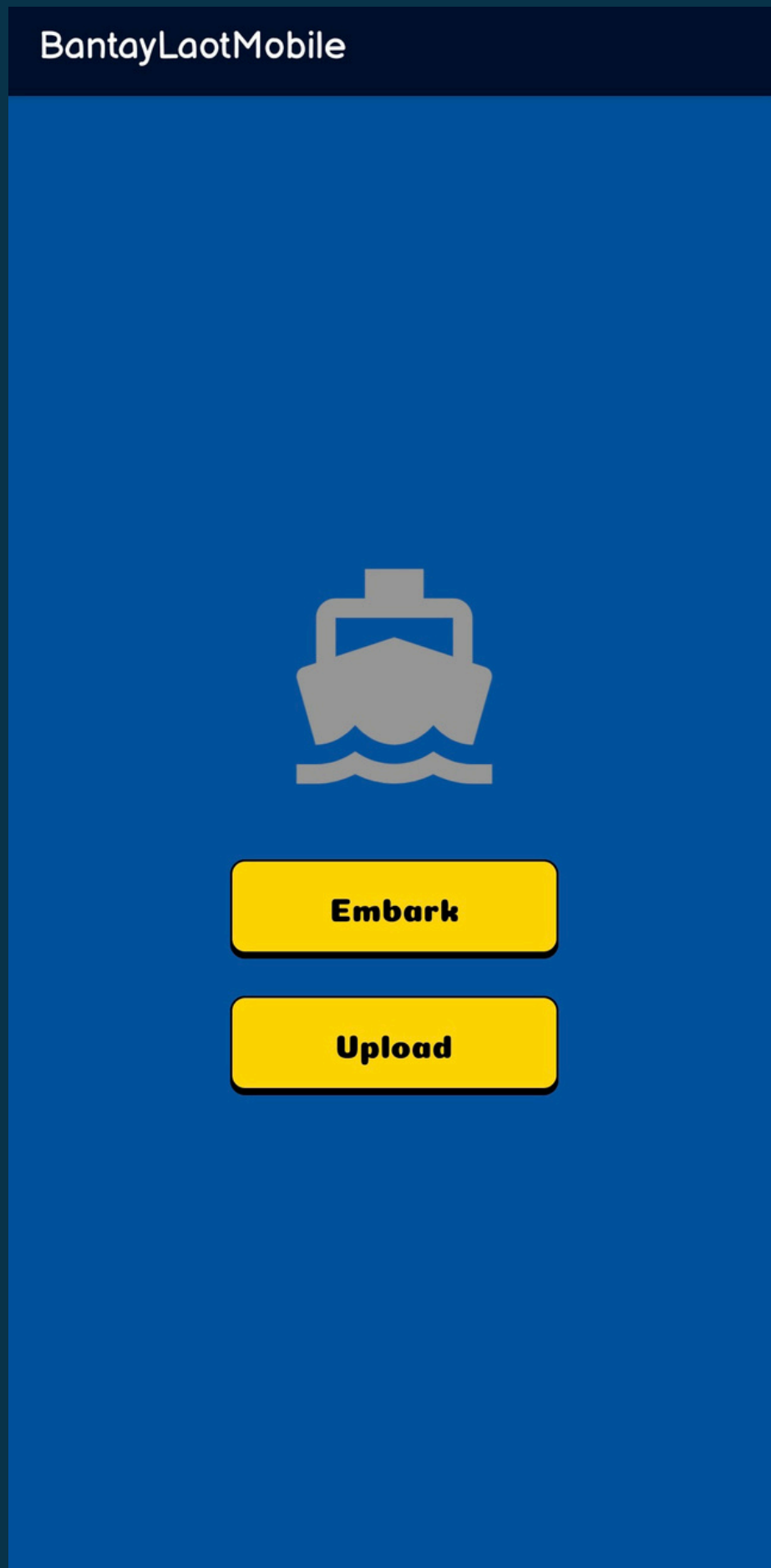
- **IUU Reporting Data:**

- Violation Details: Specific violations
- Fishing Gear Involved: Illegal or unauthorized gear
- Violator Information: Identifier of violator and vessel
- Location of Violation: GPS coordinates
- Supporting Evidence: Photos or multimedia evidence





System Features - Mobile Application



BantayLaotMobile

Fish Catch

Surface Gill Net

Bangus

24

×

Add Catch

Bottom Set Gill Net

Salmunete

5

×

Add Catch

Hook And Line

Empty

Add Catch

Submit

BantayLaotMobile

Violation Report



Juan

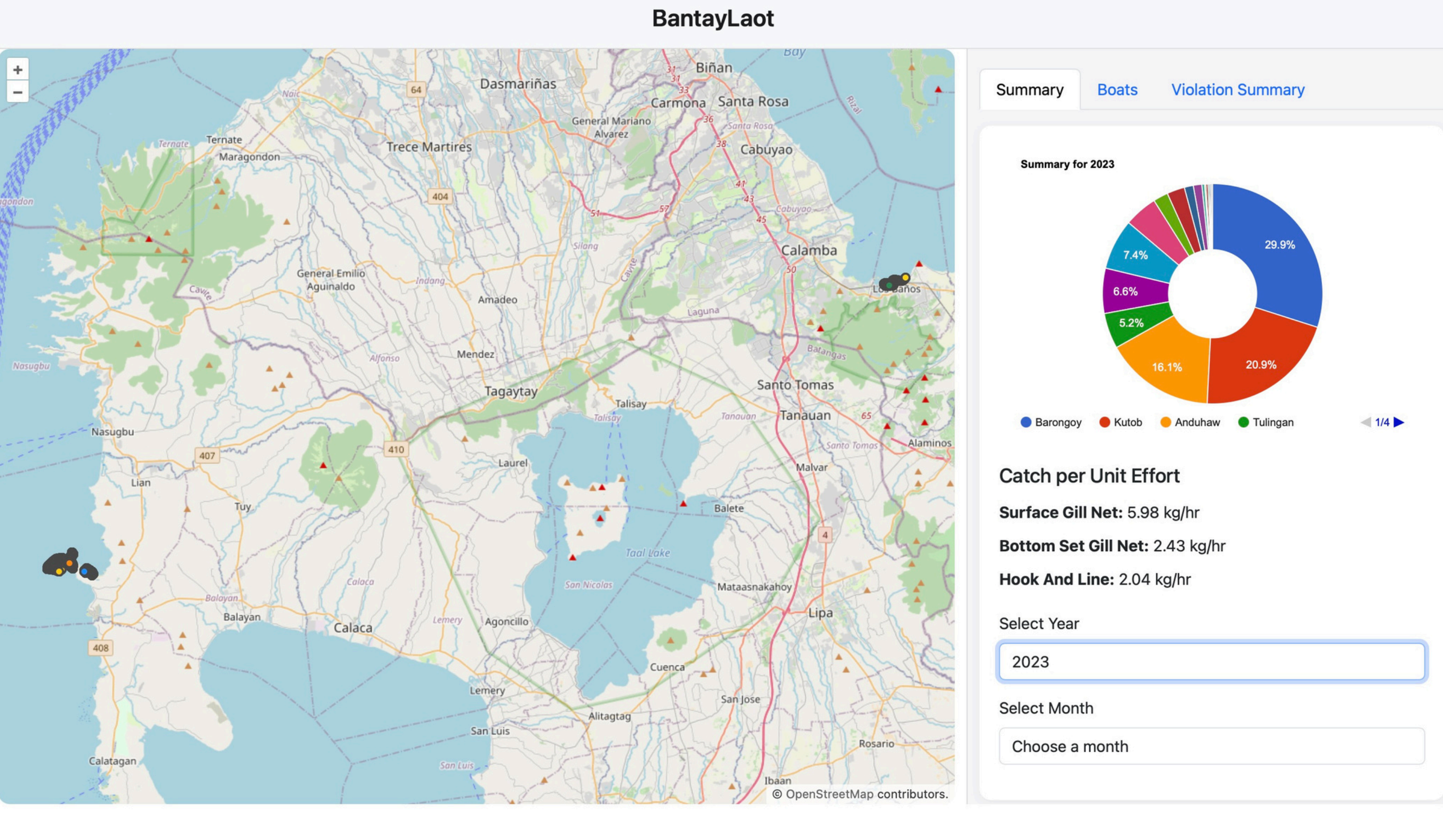
Section 126. Possessing, Dealing in or Dispo

San Jose

Dynamite fishing/ Blast fishing

143

Submit



Summary for 2023

Species	Percentage
Aduhaw	21.4%
Catfish	10.4%
Black Fath...	15.4%
Hagupit	15.4%
	6%
	11.9%
	11.4%

Aduhaw

Catfish

Black Fath...

Hagupit

1/3

Catch per Unit Effort

Surface Gill Net: 1.00 kg/hr

Bottom Set Gill Net: 1.09 kg/hr

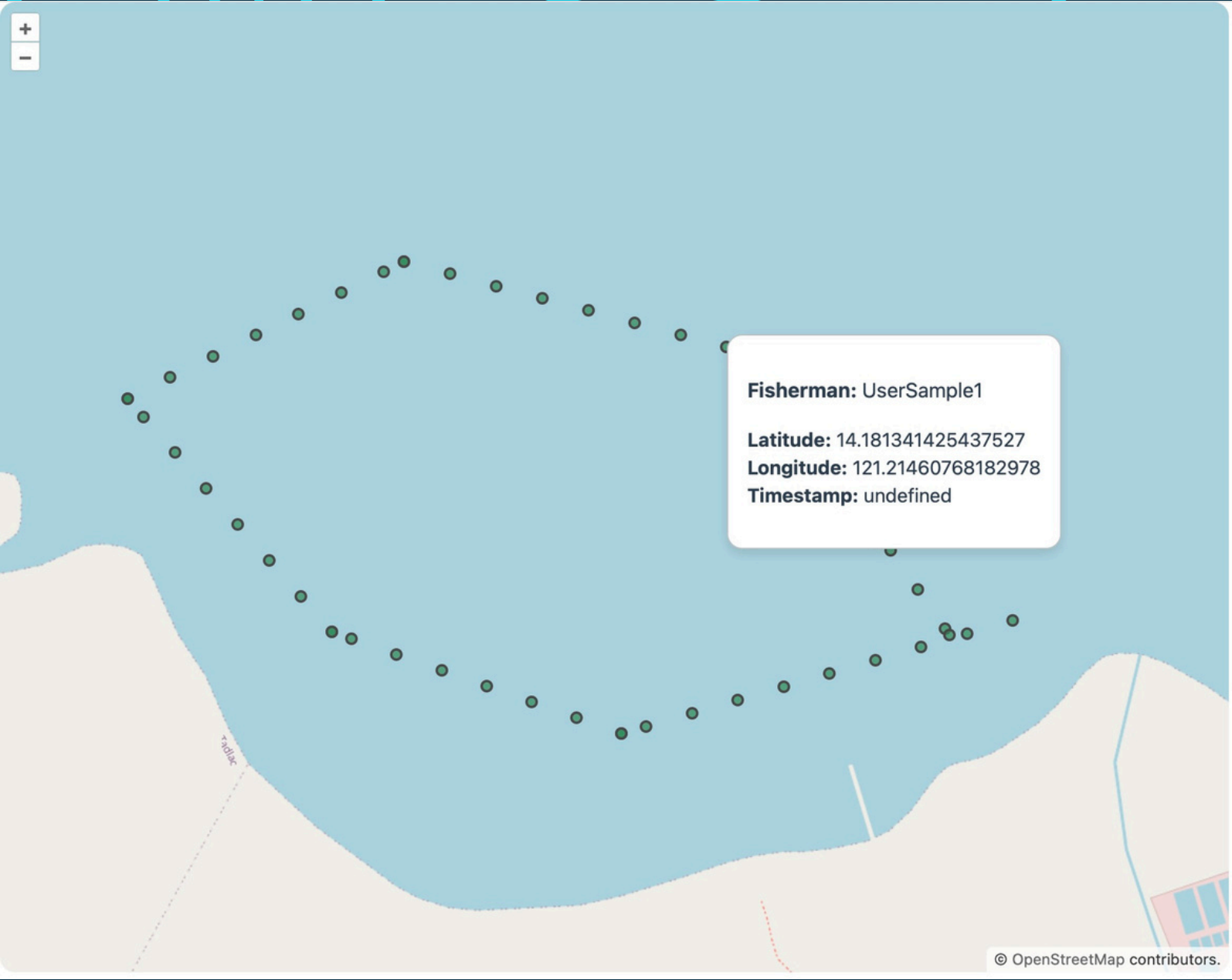
Hook And Line: 0.42 kg/hr

Select Year

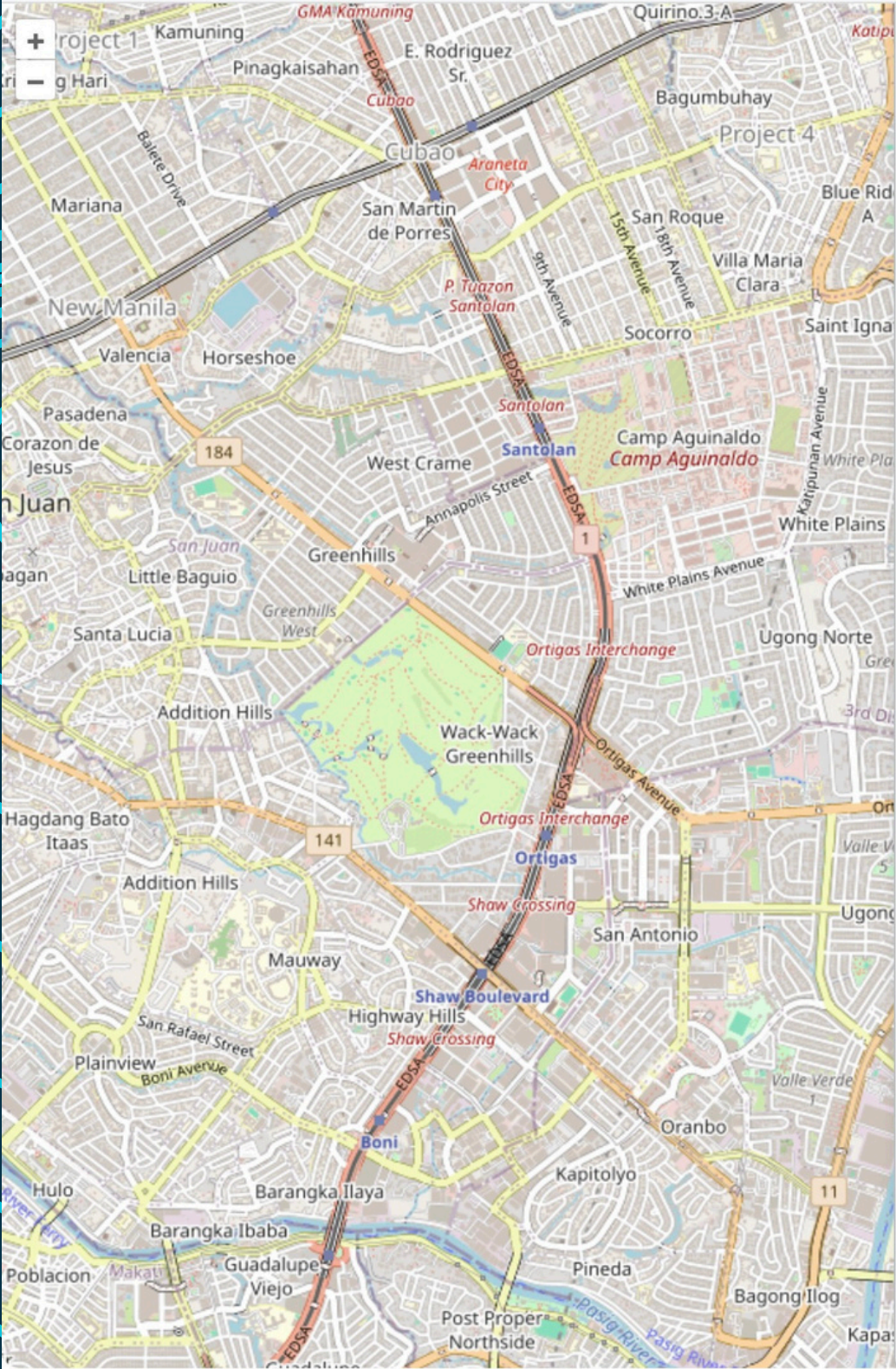
2023

Select Month

Choose a month



BantayLaot



Latitude: 14.6328217
Longitude: 121.0744643
Timestamp: 2024-11-29T20:58:28.729753
Violator: Demo

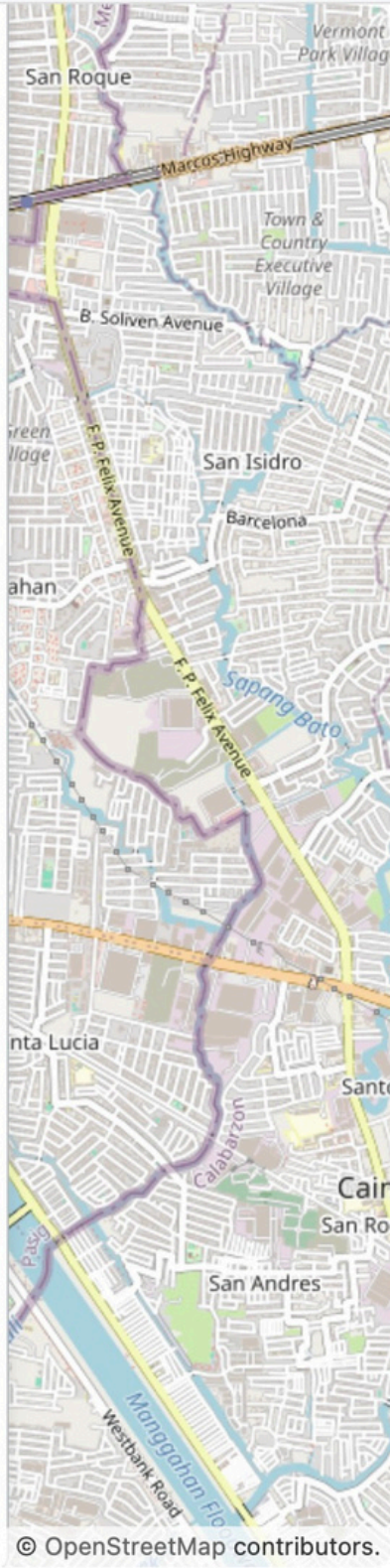
Violation: Section 123.
Concealing, Tampering or
Disposing of Evidence Relating to
an Investigation of a Violation.

Origin: Katipunan

Fishing Gear: Scoop net with
attracting device

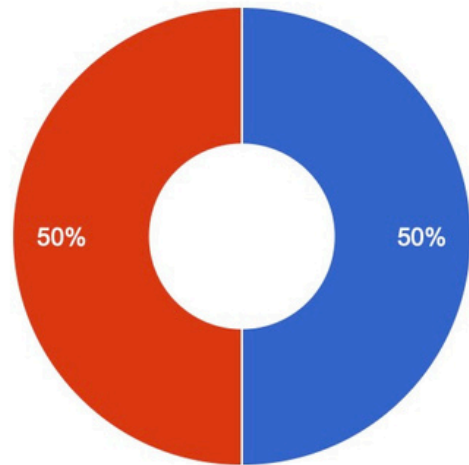
Fishing Vessel: None

Reporter: testinguser



Summary Boats Violation Summary

Violation Summary for 2024



Section 116. Noncompliance with Fisheries Observer Coverage. Section 123. Concealing, Tampering or Disposing of Evidence Relating to an Investigation of a Violation.

Total Violations: 2

Violation Reports

11/29/2024, 11:06:57 AM - Section 116. Noncompliance with Fisheries Observer Coverage. - Reporter: testinguser2

11/29/2024, 8:58:28 PM - Section 123. Concealing, Tampering or Disposing of Evidence Relating to an Investigation of a Violation. - Reporter: testinguser

Select Year

2024

Select Month

SummaryBoatsViolation Summary

Violation Summary for 2024

50%

50%

Section 116. Noncompliance wit...

Section 123. Concealing, Tampe...

Total Violations: 2

Violation Reports

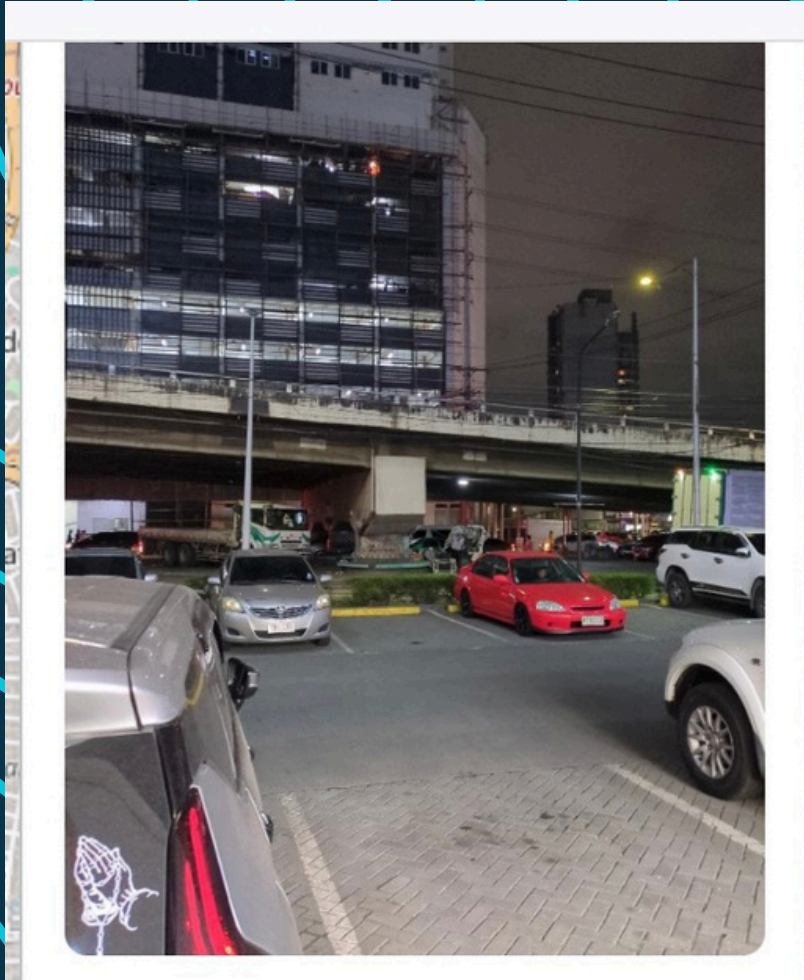
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Select Year

2024

Select Month



Latitude: 14.6328217

Longitude: 121.0744643

Timestamp: 2024-11-29T20:58:28.729753

Violator: Demo

Violation: Section 123.

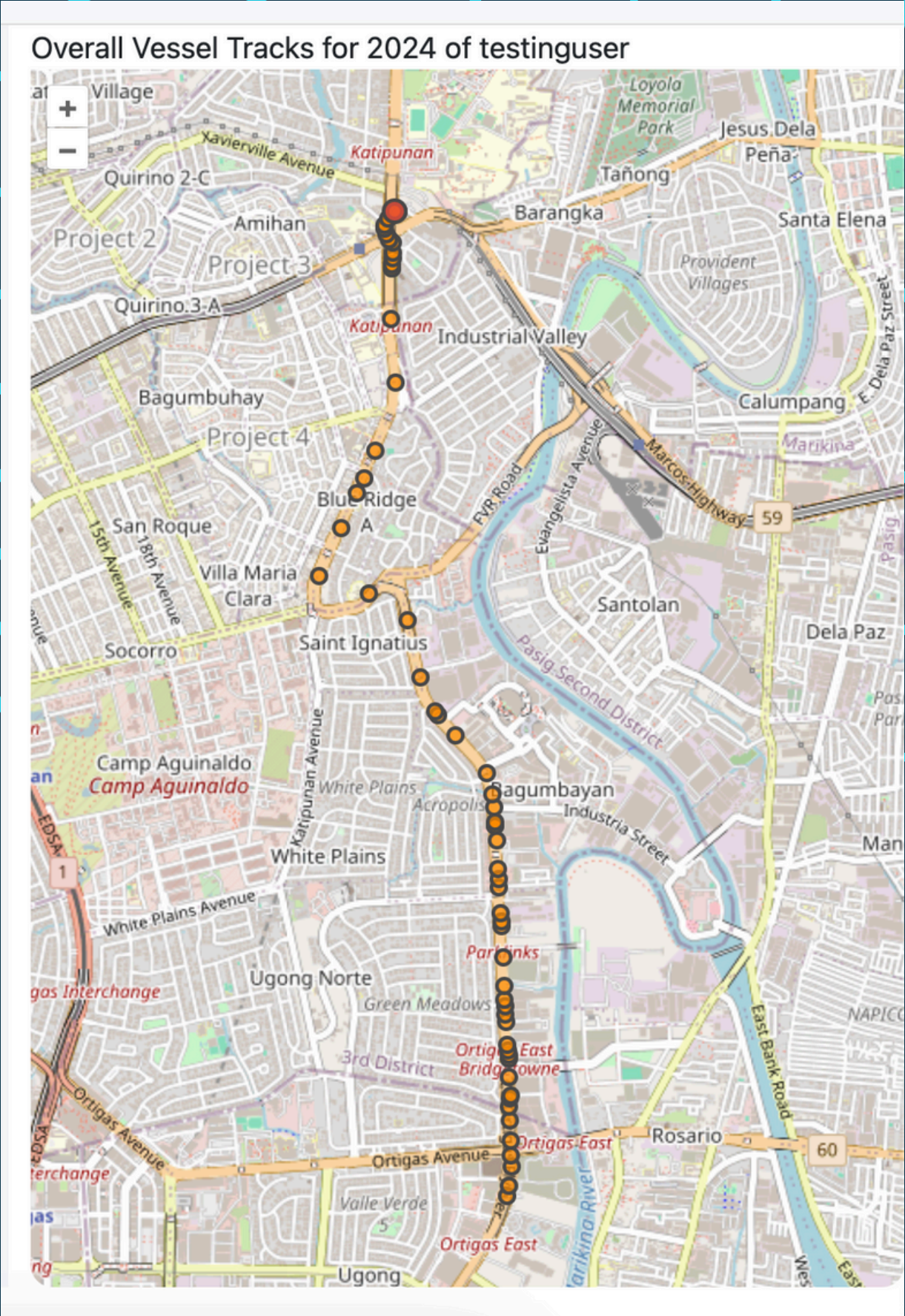
Concealing, Tampering or Disposing of Evidence Relating to an Investigation of a Violation.

Origin: Katipunan

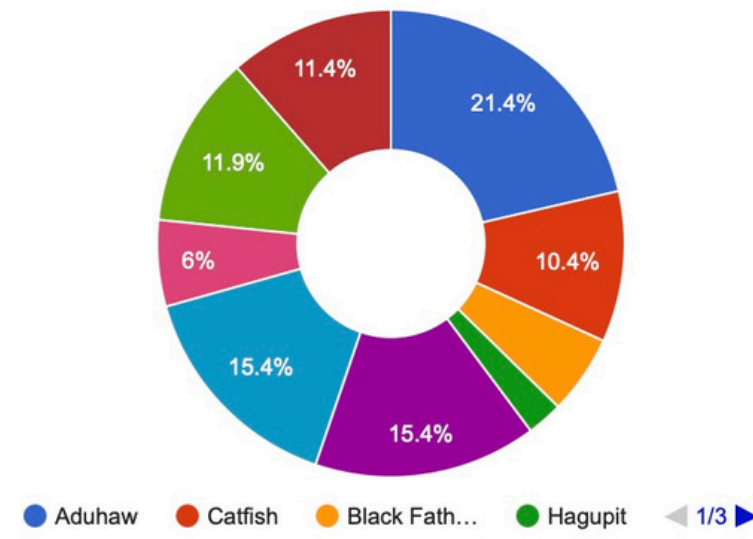
Fishing Gear: Scoop net with attracting device

Fishing Vessel: None

Reporter: testinguser



Summary for 2023



Catch per Unit Effort

Surface Gill Net: 1.00 kg/hr

Bottom Set Gill Net: 1.09 kg/hr

Hook And Line: 0.42 kg/hr

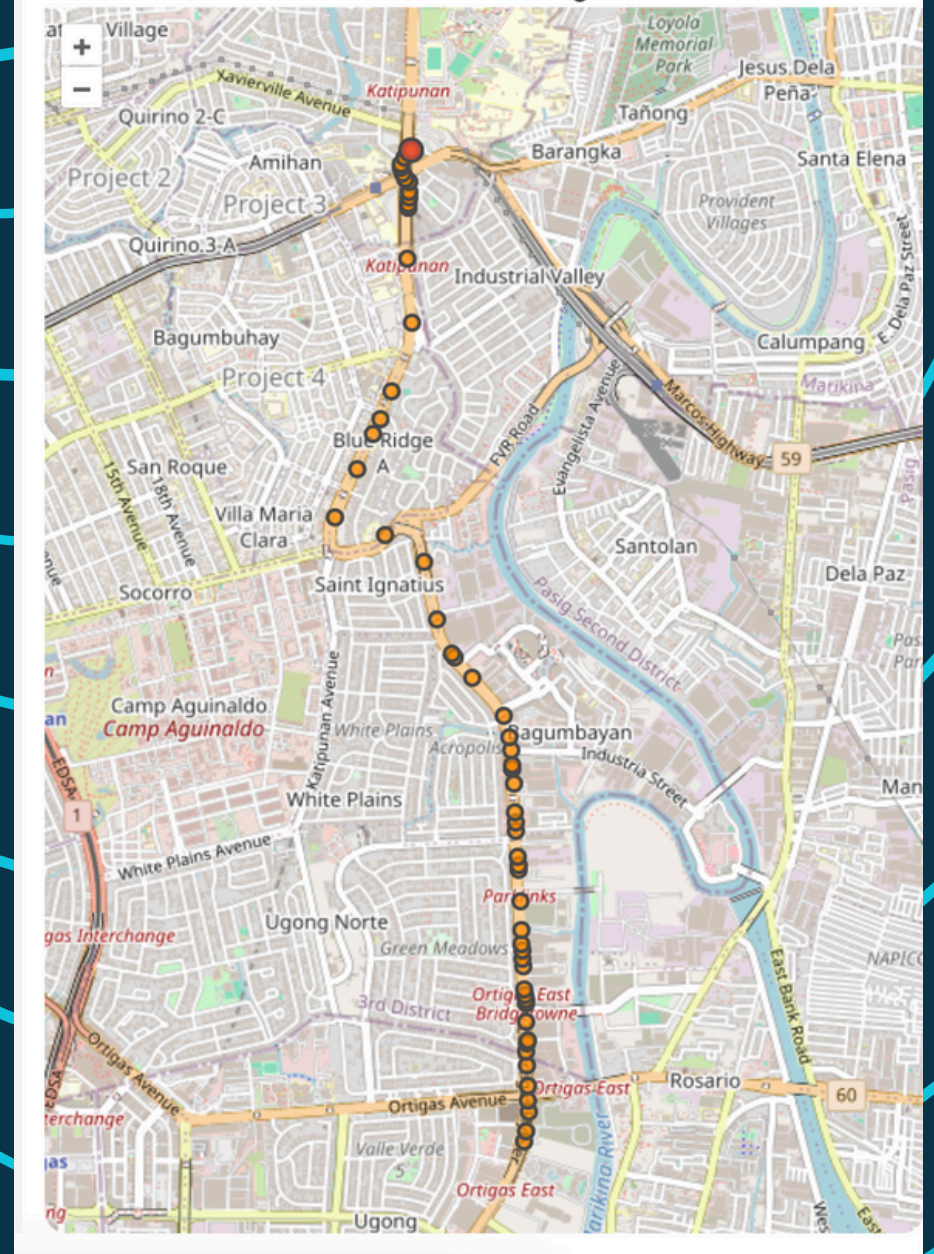
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Violation: Section 123.
Concealing, Tampering or
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an Investigation of a Violation.

Origin: Katipunan**Fishing Gear:** Scoop net with
attracting device**Fishing Vessel:** None**Reporter:** testinguser

- **Mobile App:** Data collection and synchronization
- **Web Platform:** Geospatial visualization and analytics

Overall Vessel Tracks for 2024 of testinguser



Latitude: 14.6328217
Longitude: 121.0744643
Timestamp: 2024-11-29T20:58:28.729753
Violator: Demo

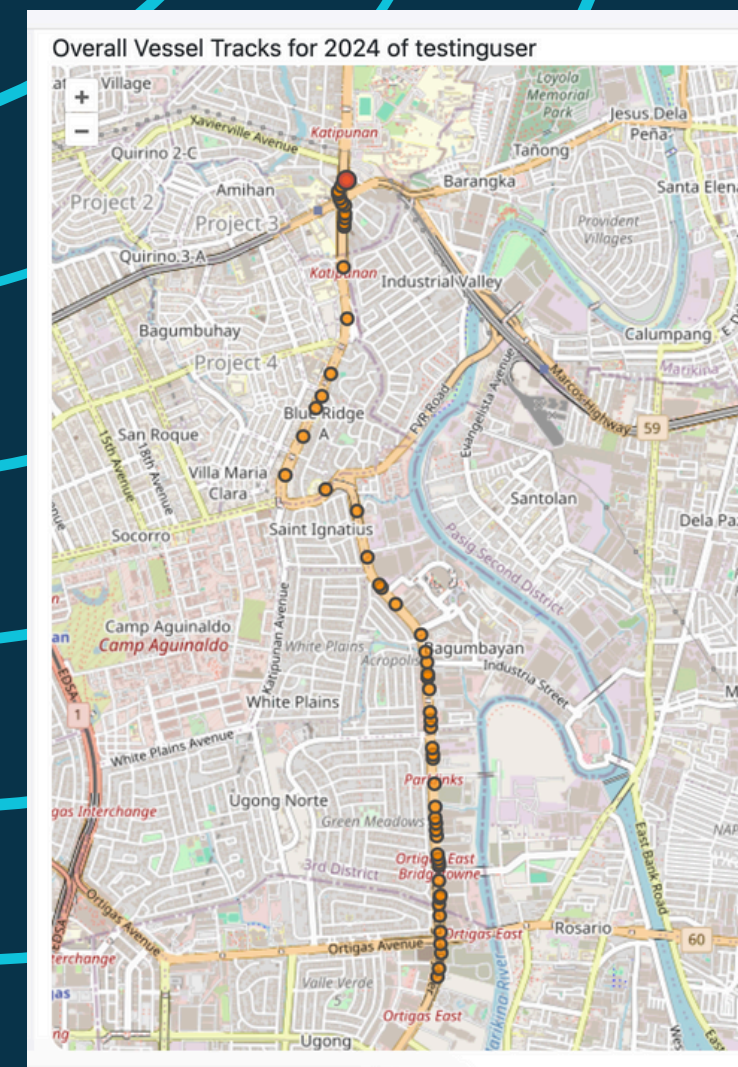
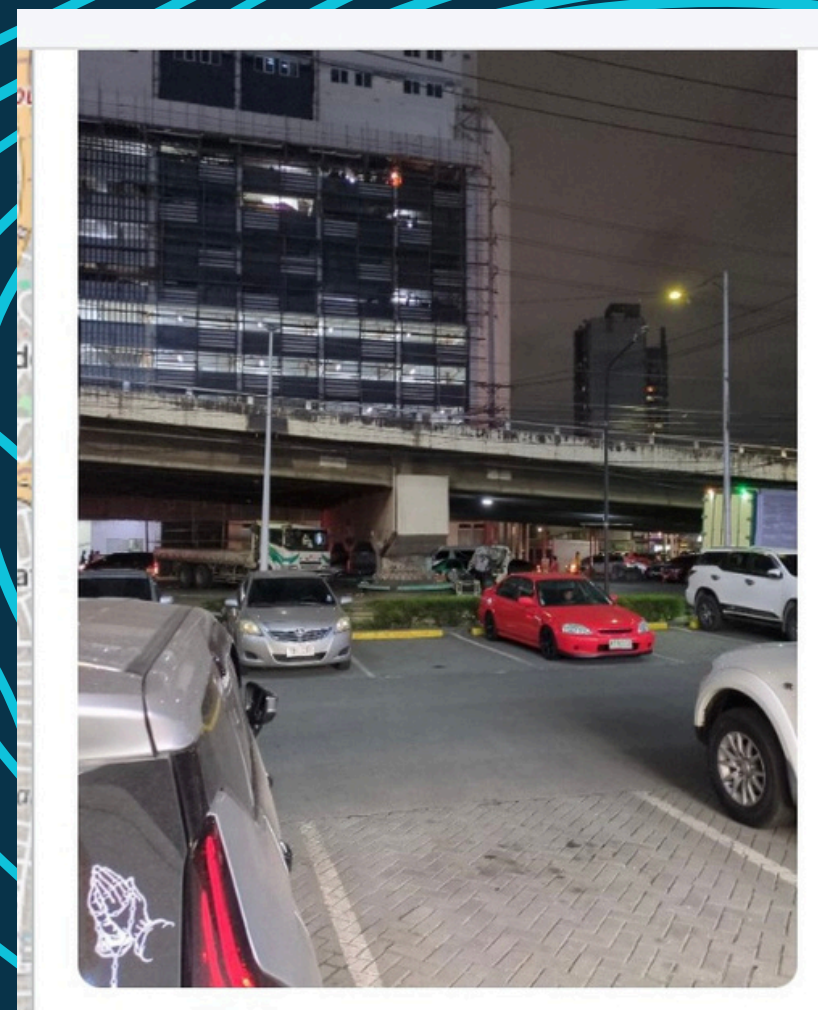
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- **Testing:** Real-time land simulations validated functionality
- **Limitations:** No field testing on actual vessels

- **Conclusion:** BantayLaot is a robust solution capable for monitoring IUU fishing
- **Recommendations:**
 - Conduct field testing
 - User account system for security
 - Image processing for automated fish recognition
 - Connect with government regulatory systems



Thank You!